01 / PROBLEM AND CONTRIBUTION

Verilog digital clock & alarm

Digital / embedded systems project | Undergraduate | Four-person team

RESULT AT A GLANCE

Implemented clock and alarm logic in Verilog and documented a ModelSim example in which the alarm asserts at the programmed 10:20 match.

The engineering question

How can sequential RTL keep time and assert an alarm when the current time matches a programmed value?

Situation

A digital clock must update time and recognize an alarm match while supporting reset and time-setting inputs. The team used a Verilog model and a simulation testbench to explore this sequential behavior.

Task

Contribute the Verilog implementation and documentation, then inspect the time counters and alarm signal in ModelSim.

My contribution

Project research, Verilog code, and report preparation; teammates also contributed circuit diagrams and testbench code.

Tools and concepts

Verilog / Sequential logic / Counters / ModelSim / RTL simulation

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

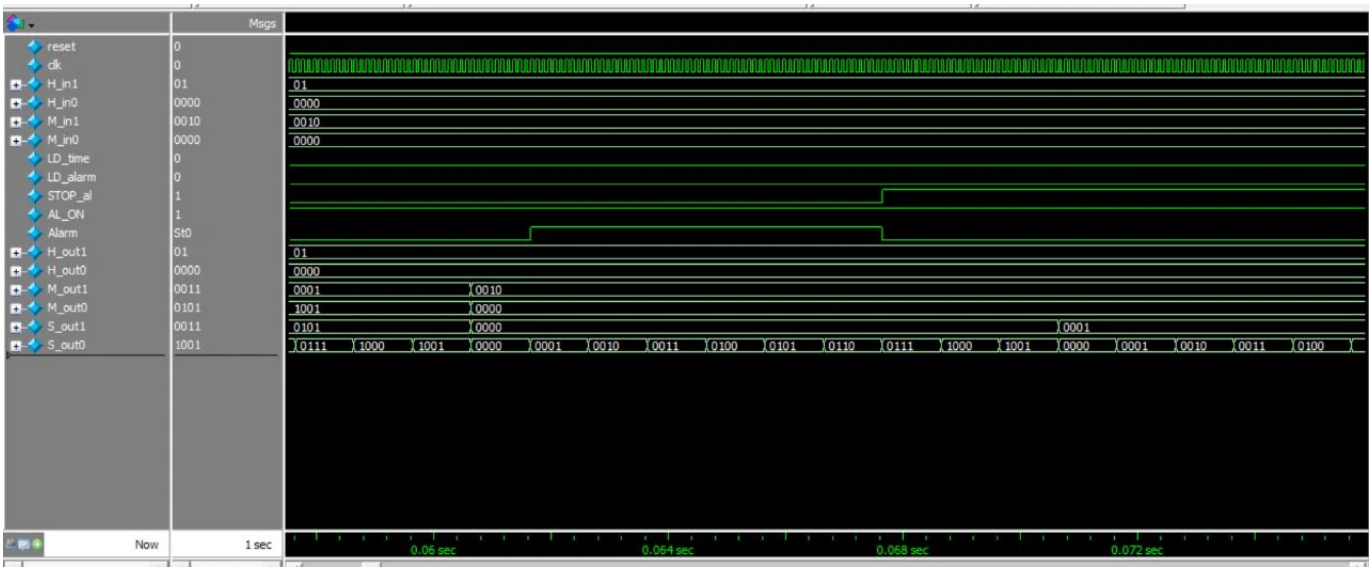
Organized the clock around counters and time-setting inputs, with reset and alarm comparison logic. The project report also includes the proposed display and control organization.

02 / Implementation step

Used ModelSim waveforms to follow the time count and the alarm output. The documented example programs an alarm for 10:20.

Sequential timekeeping

Counters represent the time state, while reset and setting inputs control how it changes. Alarm comparison observes that state against the programmed alarm; simulation allows both counter values and the alarm output to be inspected together.



Digital clock simulation

ModelSim waveform from the Verilog clock report; this is simulation evidence.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Reported result

The report describes time advancing through the day and the alarm output asserting at the programmed match. Verilog and testbench excerpts accompany the waveform.

What the example demonstrates

The documented 10:20 alarm match is a functional simulation example. It does not establish oscillator accuracy, FPGA resource usage, board-level display behavior, or implementation timing closure.

Scope and limitations

The available evidence is a simulation report. A completed FPGA board deployment and timing closure are not claimed here.

Possible next steps - proposed, not completed results

- Add directed rollover, reset, and alarm-reprogramming tests with expected outcomes.
- If deployed to a board, document clock division, pin constraints, and measured behavior separately from simulation.

Source material: FPGA-Based Digital Clock Design Using Verilog.docx, results, contribution list, and appendix.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.